

Helping Faculty Integrate GenAI into CS Education

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**GenAI in CS
Education**
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The Software Engineering (SWE) Industry is Using GenAI

- 92% of US-based developers report using GenAI [1]
- At Microsoft, 20-30% of code is written by GenAI [2]
- A quarter of recent startups have code bases where >95% of their code comes from AI [3]
- Industry interviews are now requiring GenAI use [4]

1. <https://github.blog/2023-06-13-survey-reveals-ais-impact-on-the-developer-experience/>
2. <https://www.cnbc.com/2025/04/29/satya-nadella-says-as-much-as-30percent-of-microsoft-code-is-written-by-ai.html>
3. <https://techcrunch.com/2025/03/06/a-quarter-of-startups-in-yocs-current-cohort-have-codebases-that-are-almost-entirely-ai-generated/>
4. <https://www.hellointerview.com/blog/meta-ai-enabled-coding>

Skills to Succeed in the Modern SWE Workplace

Research led by Google studying how leading developers are using AI [1] found that successful AI-enhanced developers have skills in:

1. **Using Generative AI effectively** ← AI skills are necessary
 - e.g. assessing AI outputs, calibrating interactions with AI
2. **Core software engineering**
 - Coding, testing, risk assessment, system design... ← SWE “core” is still there!
3. **Adjacent engineering**
 - Cybersecurity, regulations, technological trends
4. **Adjacent non-engineering**
 - e.g. interacting with end users, awareness of competitor products, knowledge of market trends

[1] Kam, M. et al. (2025). What do professional software developers need to know to succeed in an age of Artificial Intelligence?. In Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering.



AI in the Workforce Report

DEC Partnered with the Global Finance & Technology Network (GFTN)

Employer Expectations:

51% Employers expect new graduates to be proficient in using AI tools.

53% are worried about graduates' ability to critically evaluate and build upon AI-generated content.

Perceived Lack of Preparation:

3% of employers believe that institutions are adequately preparing graduates for an AI-driven workforce

80% of employers feel that higher education is not keeping up with the pace of industry change

Most Valued Skills:

92% Critical and Analytical Thinking

62% Ability to Work with AI

→ Over domain expertise (19%)

Debate within Academia:

Faculty views I've heard this year (paraphrased):

“Students should only use AI to do things they can do themselves from scratch first.”

“A student using AI in any of our courses is the same as contract cheating.”

“Engineers using GenAI in industry were trained in the classic curriculum. Until I see evidence to the contrary, we shouldn't change anything.”

“If we don't change our curriculum we'll become as relevant to modern software design as learning ancient Greek to read the Iliad is to modern creative writing.”

Not a New Problem - Just a More Pressing One

- Decades of research have shown a large divide between what is taught in CS courses and the skills needed in industry [1]
 - Well-defined assignments vs. open-ended project requirements
 - Short time span on assignments vs. working with legacy code
 - Working individually vs. working on a team
 - Writing standalone programs vs. adding features to codebases

“AI makes old problems urgent” - Titus Winters, Adobe

Many Faculty Hope to Close the Academia/Industry Preparation Gap

- Encouragingly, many faculty want to help prepare students for industry and are willing to improve their curriculum to do so [1]
- But they face barriers to change:
 - Industry skills not in the curriculum/courses they teach
 - Creating meaningful assignments can be difficult
 - Lack of recent industry experience

1. Valstar, Sih, Krause-Levy, Porter, and Griswold. A Quantitative Study of Faculty Views on the Goals of an Undergraduate CS Program and Preparing Students for Industry. ICER 2020.

Many Educators Recognize the Need for Change

- Educators believe the skills required to program have changed
 - 75% of educators acknowledge that the skills to program are changing because of GenAI
 - But... only 30% of educators are integrating it into their classes



Consortium for GenAI in CS Education

- Goal is to **support educators** with integrating **GenAI** into the **CS** curriculum
- Consortium housed by UCSD is co-run by Leo, Dan Zingaro, and Beth Simon
 - Funded by Google.org
- Support consortium partners:
 - CRA, SIGCSE, CSTA, ADMI, CAHSI, Raspberry PI + more
- Host 7 courses with materials for educators to adopt:
 - CS1 (10 week), CS1 (15 week), CS2, Algorithms, Software Engineering, and Product Management, Navigating Large Code Bases



www.teachCSwithAI.org



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Consortium Goals

- **Outreach:** Sharing resources with the community
- **Resource Development:** Creating courses, assessments, tailored materials for partner communities
- **Community of Practice:** Support partner members through communities of practice



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Attributes of the shared courses

- **Course learning goals updated to reflect the skills needed to succeed when working with GenAI tools**
 - E.g., less emphasis on code writing from scratch, more on specification, verification, and debugging
- **Redesigned Assessments**
 - Students use GenAI tools to complete take-home projects

Two lessons I learned in Day 0

Software creation with GenAI has changed more than I'd realized and the skills to write software with these new tools are still a bit unclear

**We need close academia/industry
collaborations more than ever**

Thank you to our many collaborators!



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Learn more about the Consortium:

www.teachCSwithAI.org

Materials for CS1, CS2, Algorithms, SWE, Product Management, and Large Code Bases courses:

<https://www.teachcswithai.org/courses>

